

COLLEGE ADMISSION → ENROLLMENT → GRADUATION → TOR Comprehensive Development Guide

A practical blueprint for designing a college student lifecycle, academic records, enrollment, grading, curriculum, graduation, and Transcript of Records system.

Scope: Admission, enrollment, curriculum, course/section assignment, advising, grades, academic standing, financial/administrative controls, internship/OJT, research, graduation, registrar records, TOR issuance, post-graduation, and alumni.

Development principle: the system should preserve the student's complete academic history while clearly separating official registrar records from planning/working records.

1. Document Purpose and Scope

This guide converts the complete college lifecycle into a development-oriented blueprint. It can be used as a starting point for requirements gathering, database design, workflow design, UI/UX planning, API development, testing, and registrar operations.

Primary lifecycle:

Pre-admission → Application → Admission → Confirmation → Enrollment → Advising → Course/section assignment → Semester delivery → Assessment → Grades → Academic standing → Re-enrollment → Graduation audit → Clearance → Degree conferment → TOR/certifications → Licensure/employment/further study → Alumni.

2. Design Principles

- Official-source principle: the registrar/authorized academic record is authoritative for official enrollment, grades, credits, and degree completion.
- Historical-record principle: do not overwrite meaningful academic history; preserve previous attempts, status changes, curriculum versions, and approved transactions.
- Workflow principle: important changes should move through request → validation → approval → posting → audit trail.
- Curriculum principle: bind students to the correct curriculum version and support approved transitions.
- Security principle: use role-based access, authentication, least privilege, audit logs, backups, and controlled document release.
- Configurability: grading scales, units, prerequisites, statuses, honors rules, and document requirements should be configurable rather than hard-coded.
- Privacy: collect only necessary data, restrict access, and follow the institution's applicable privacy and records-retention policies.

3. Major System Modules

Module	Primary responsibility
1. Student Master Record	Identity, student number, program, cohort, status, contacts
2. Admissions	Applications, requirements, exams, interviews, decisions
3. Document Management	Uploaded/submitted documents, verification, expiry, status
4. Program & Curriculum	Programs, curriculum versions, courses, prerequisites, substitutions
5. Academic Advising	Degree planning, course recommendations, approvals
6. Enrollment	Term registration, course selection, add/drop, withdrawal
7. Sections & Scheduling	Sections, instructors, rooms, schedules, capacity/conflicts
8. Assessment & Grades	Assessment components, grades, grade approval, posting
9. Academic Standing	GPA, deficiencies, probation/other statuses per policy
10. Finance/Scholarships	Charges, payments, holds, aid, scholarship monitoring
11. Student Services	ID, library, health/clearance, support services
12. Internship/OJT	Placement, hours, attendance, evaluations, completion
13. Research/Thesis/Capstone	Proposal, adviser, panel, defense, final submission
14. Graduation	Application, degree audit, clearance, approval, conferment

Module	Primary responsibility
15. Registrar	Official records, certifications, corrections, verification
16. TOR & Credentials	Transcript generation, requests, release, verification
17. Alumni/Post-Graduation	Alumni status and optional post-graduation records
18. Security & Audit	Roles, permissions, logs, backups, data controls

4. End-to-End Student Lifecycle

Use the following as the master workflow for the application.

PRE-COLLEGE → STUDENT PROFILE → UNIVERSITY/PROGRAM SELECTION → APPLICATION → DOCUMENT VERIFICATION → EXAM/INTERVIEW → ADMISSION DECISION → ACCEPTANCE → STUDENT NUMBER/PORTAL → ENROLLMENT → CURRICULUM ASSIGNMENT → ADVISING → COURSE SELECTION → PREREQUISITE CHECK → SECTION ASSIGNMENT → SCHEDULE CONFIRMATION → ASSESSMENT/PAYMENT → OFFICIAL REGISTRATION → SEMESTER → ATTENDANCE/ASSESSMENTS → MIDTERM → FINAL GRADES → ACADEMIC STANDING → SEMESTER AUDIT → NEXT TERM → YEARLY CURRICULUM AUDIT → INTERNSHIP/OJT → RESEARCH/THESIS/CAPSTONE → GRADUATION APPLICATION → DEGREE AUDIT → CLEARANCE → GRADUATION APPROVAL → DEGREE CONFERMENT → TOR/DIPLOMA/CERTIFICATIONS → LICENSURE/EMPLOYMENT/FURTHER STUDY → ALUMNI.

5. Student Master Record

The Student Master Record should contain the stable identity and institutional identifiers. Avoid storing calculated or temporary information here when it belongs in transaction tables.

Field group	Examples
Identity	Student ID, legal name, preferred name, date/place of birth
Contact	Email, mobile, address, emergency contact
Academic identity	Program, major, campus, college, cohort, curriculum version
Admission	Admission type, admission term, application reference
Status	Current student status, effective date, status history
Accounts	Portal account reference, university email reference
Audit	Created by/date, updated by/date, record version

6. Admissions Module

Admission workflow:

Draft → Submitted → Document Check → Entrance Exam/Interview → Evaluation → Accepted / Waitlisted / Denied → Confirmation → Enrollment eligibility.

Track:

- Application number and date
- Applicant category: freshman, transferee, returning, second-degree, international/non-degree where applicable
- Preferred program and alternatives
- Admission requirements and verification status
- Entrance exam and interview records
- Admission decision and conditions
- Acceptance deadline and enrollment deadline
- Scholarship/financial-aid applications
- Communication and follow-up history

7. Admissions Document Management

Documents should have their own status rather than simply being attached to an application.

Document	Suggested statuses
Birth/civil document	Required → Submitted → Verified → Rejected/Needs correction
School record	Required → Submitted → Verified
Good moral/certificate	Required → Submitted → Verified
ID/photo	Required → Submitted → Verified
Medical/other requirement	Required → Submitted → Verified
Scholarship documents	Required → Submitted → Evaluated → Approved/Denied

8. Transfer Credit & Advanced Standing

For transferees and students with prior coursework, maintain an equivalency table rather than replacing the original course history.

Previous course	Units	Grade	Equivalent	Credit decision	Approver
Previous course A	3	1.75	Course X	Credited	Registrar/Dept.
Previous course B	3	2.00	Course Y	Not credited	Registrar/Dept.

Also support residency rules, course equivalency evidence, official transfer documents, and approved credit decisions.

9. Program & Curriculum Management

A curriculum is more than a list of subjects. Model its version, effective academic year, required courses, electives, prerequisites, credits/units, and completion rules.

Entity	Important data
Program	Program code, title, department, degree, duration
Curriculum version	Version/code, effective term/year, status
Course	Code, title, units, type, description
Curriculum course	Required/elective, year/term placement, sequence
Prerequisite	Course relationship, condition
Corequisite	Course relationship, condition
Substitution	Required course, substitute, approval, effective date
Transition rule	Old curriculum, new curriculum, cohort/conditions

10. Academic Advising & Degree Audit

The degree audit should answer: What has the student completed, what is currently enrolled, what remains, and what is permitted next?

Recommended audit categories:

- General education requirements
- Major/core requirements
- Electives
- PE/NSTP or institution-specific requirements
- Internship/OJT/practicum
- Research/thesis/capstone
- Free electives where applicable
- Minimum total units
- Minimum residency requirements
- Other graduation conditions

11. Enrollment Workflow

Enrollment should be treated as a controlled transaction rather than simply adding subjects to a student's record.

Term setup → student eligibility check → advising → course selection → prerequisite validation → section/capacity check → schedule conflict check → approval → assessment → payment/financial clearance → official registration → registration document/COR.

12. Section & Schedule Management

Field	Examples
Section	BSCS-1A

Field	Examples
Course	Programming 1
Instructor	Instructor record
Room	Room/Lab
Schedule	Day/time
Capacity	Maximum seats
Enrollment	Current seats
Modality	Face-to-face/hybrid/online, if applicable

13. Add/Drop/Withdrawal

Every enrollment change should record the request, reason, effective date, approval, resulting schedule, and academic/financial consequences.

Supported transactions may include:

- Add course
- Drop course
- Change section
- Change schedule
- Course withdrawal
- Administrative removal
- Approved course substitution

14. Student Status and Exceptional Paths

Event	Workflow
Leave of absence	Request → review → approval → effective term → return/readmission
Readmission	Application → academic/administrative checks → approval → enrollment
Shifting	Request → evaluation → new curriculum → credited courses → approval
Transfer out	Request → clearance → official records/document release
Program change	Request → eligibility → curriculum update → approval
Dismissal/disciplinary action	Policy-based process → decision → appeal where applicable

15. Assessment & Grade Management

Separate assessment components from the official final course grade.

Layer	Examples
Assessment	Quiz, exam, project, laboratory, assignment, participation
Period	Prelim/midterm/final or institution-defined periods
Final grade	Official course result
Status	Passed, failed, incomplete, withdrawn, deferred, etc., per policy
Approval	Instructor → department/authorized approval → registrar posting

16. Gradebook and Academic Record

Maintain a working gradebook for teaching/monitoring and an official academic record for registrar purposes. Do not let an editable classroom gradebook silently alter an official posted grade.

17. GPA / Academic Standing

Where the institution uses a unit-weighted GPA, a generic formula is:

$$\text{GPA} = \Sigma(\text{Grade} \times \text{Units}) \div \Sigma \text{Units}$$

The actual grading scale, rounding rules, repeat-course treatment, honors rules, and academic-standing thresholds must come from the institution's approved policies rather than being hard-coded into the system.

18. Academic Exceptions

Exception	System requirement
Incomplete	Track incomplete status, deadline, completion, and final resolution
Failed	Preserve failed attempt and support repeat/retake according to policy
Repeated course	Preserve every attempt; apply institutional GPA/TOR rules
Missing grade	Flag before term closure
Grade correction	Request → review → approval → audit log → official update
Academic misconduct	Separate disciplinary workflow; preserve decision and audit trail

19. Finance, Holds & Scholarships

Enrollment eligibility may depend on administrative or financial conditions. Keep financial transactions separate from academic grades, while allowing controlled eligibility checks.

Area	Records
Charges	Tuition, miscellaneous, laboratory and other institutional charges
Payments	Amount, date, reference, method
Balance	Outstanding amount/status
Holds	Type, reason, effective date, release date
Scholarship	Application, award, conditions, renewal, termination

20. Student Services

Support services can include student ID, library, health/medical clearance where required, student organizations, counseling/support, career services, and portal/email accounts. These should integrate through controlled identifiers rather than duplicating the student master record.

21. Internship / OJT / Practicum

Stage	Records
Eligibility	Prerequisites and approved term
Placement	Host organization, supervisor, agreement
Execution	Start/end dates, required hours, attendance
Evaluation	Student/host/institution evaluations
Completion	Verified hours, completion certificate, academic result

22. Research / Thesis / Capstone

Recommended workflow: Topic → adviser → proposal → approval → research/development → defense → revisions → final submission → completion.

Track title, adviser, panel, proposal approval, defense date, result, revision status, final submission, and repository/document reference where applicable.

23. Graduation Application & Degree Audit

Graduation should be driven by an auditable degree-completion process, not simply by the number of years attended.

Student application → registrar/college degree audit → curriculum requirements check → remaining deficiencies → internship/research verification → residency/unit check → financial/library/department clearances → graduation approval.

Requirement	Audit question
Curriculum	Has every required course/category been satisfied?
Units	Are required total units completed?
Prerequisites	Were courses completed in an acceptable sequence?
Internship/OJT	Is the requirement complete and documented?
Thesis/capstone	Is the requirement approved and complete?
Residency	Are institutional residency requirements satisfied?
Clearance	Are required administrative clearances complete?

24. Graduation & Degree Conferment

Record the approved graduation date, degree, major, college/department, honors or distinctions where applicable, and degree-conferment status.

25. Registrar Master Record

The registrar layer should be the authoritative source for official academic history and credential generation. It should be protected from unauthorized edits and maintain a complete audit history.

Record	Examples
Identity	Student identity and institutional identifiers
Enrollment	Official term registrations and registration status
Academic history	Courses, units, grades, repeats, withdrawals
Program history	Program changes and curriculum versions
Academic standing	Official status/history where maintained
Completion	Degree audit and graduation status
Credentials	TOR, certifications, diploma references
Corrections	Approved changes with reason, evidence, actor, timestamp

26. Transcript of Records (TOR) Workflow

Recommended lifecycle:

Request → identity verification → eligibility/hold check where applicable → registrar processing → academic record assembly → validation → official generation → authentication/signature/security controls → release → request/release log.

27. TOR Data Structure

Section	Potential content
Student identification	Name, student number, other institution-approved identifiers
Program	Degree/program, major, campus/college where applicable
Academic periods	Term/semester and academic year
Course history	Course code, title, units/credits, grade
Academic totals	Institution-defined totals/GPA or equivalent information
Completion	Degree/graduation information where applicable
Authentication	Registrar authorization, document number, security/verification reference

28. TOR Corrections & Verification

Provide a controlled process for discrepancies: student/requester identifies issue → registrar investigates → supporting records reviewed → correction approved if warranted → official record updated → audit trail preserved → replacement credential issued if required.

29. Post-Graduation

The system may continue with licensure/board-exam tracking where relevant, employment/graduate-school document requests, alumni records, and credential verification.

30. Alumni

Maintain only the alumni information the institution is authorized and needs to retain. Alumni records should not overwrite the historical academic record.

31. Suggested Database Architecture

A relational database is appropriate for the structured academic lifecycle. The following is a conceptual starting point, not a fixed implementation schema.

Table/entity	Purpose
students	Master student identity
student_contacts	Contact and emergency details
applications	Admission applications
application_requirements	Requirement checklist and verification
admission_decisions	Admission outcomes
programs	Degree programs
curriculum_versions	Cohort/curriculum versions
courses	Course catalog
curriculum_courses	Courses required by curriculum
course_prerequisites	Prerequisite relationships
course_substitutions	Approved substitutions
terms	Academic terms
sections	Course sections
section_schedules	Meeting schedules
instructors	Instructor records
enrollments	Official term enrollment
enrollment_items	Courses registered in an enrollment
attendance	Attendance where applicable
assessments	Assessment components
grades	Official/working grade records with status
grade_changes	Controlled grade corrections
academic_standing	Standing history
transfer_credits	Evaluated prior coursework
program_changes	Shifting/program changes
financial_accounts	Charges/balances
payments	Payments
holds	Administrative/financial holds
scholarships	Scholarship records
internships	OJT/practicum placements
research_projects	Thesis/capstone
graduation_applications	Graduation requests
degree_audits	Requirement completion
clearances	Graduation/administrative clearances
degrees_conferred	Degree completion/conferment

Table/entity	Purpose
tor_requests	TOR requests
credentials	Issued credentials/certifications
audit_logs	System activity/history
users_roles_permissions	Access control

32. Key Relationships

Student 1→many Applications; Student 1→many Enrollments; Enrollment 1→many Enrollment Items; Enrollment Item → Section → Course; Student → Program/Curriculum Version; Course → Prerequisites; Student → Degree Audit → Graduation Application → Degree Conferment → Credential/TOR Request.

33. Status Model

Domain	Example statuses
Applicant	Draft, Submitted, Verified, Accepted, Waitlisted, Denied
Student	Admitted, Enrolled, Regular, Irregular, Leave, Withdrawn, Graduated
Course enrollment	Registered, Dropped, Withdrawn, Completed
Grade	Pending, Posted, Incomplete, Failed, Corrected
Graduation	Applied, Under Audit, Deficient, Cleared, Approved, Graduated
Credential	Requested, Processing, Released, Verified

34. Roles & Permissions

Use role-based access control. Example roles:

- Applicant/student: own application, enrollment planning, requests, and permitted personal records
- Instructor: assigned sections, attendance, assessments, working grades, grade submission
- Adviser: degree audit, advising, course recommendations
- Department/college: section/curriculum approvals and academic workflows
- Finance: charges, payments, financial holds
- Registrar: official academic records, enrollment posting, grade posting, credentials
- Admissions: application and admission processing
- System administrator: technical configuration; should not automatically have unrestricted academic editing authority
- Auditor/compliance role: read-only access to approved audit information

35. Core Business Rules

- 1 A student cannot be officially enrolled without satisfying the institution's configured enrollment eligibility rules.
- 2 A course requiring a prerequisite should be blocked or flagged unless an approved exception exists.
- 3 A section cannot exceed configured capacity unless an authorized override is recorded.
- 4 Schedule conflicts should be detected before official registration.
- 5 Official grades should not be silently overwritten.
- 6 Grade corrections must preserve the original value, reason, actor, approval, and timestamp.
- 7 Curriculum audits must use the student's applicable curriculum version.
- 8 Graduation approval must require all configured completion conditions to be satisfied.
- 9 TOR generation must use authoritative academic records rather than editable planning data.
- 10 Credential releases should be logged.
- 11 Access to personal and academic records must be limited by role and legitimate institutional need.

36. Validation & Testing Plan

Test the system by lifecycle stage and by exceptional path—not only by successful enrollment.

Test area	Examples
Admissions	Incomplete application, duplicate applicant, late submission
Documents	Wrong document, rejected document, replacement
Enrollment	Prerequisite missing, full section, schedule conflict
Course changes	Add/drop/withdrawal deadlines and approvals
Grades	Missing grade, incomplete, correction, repeat
Curriculum	Version change, substitution, program shift
Transfer	Credit evaluation and non-creditable subjects
Graduation	Missing unit, missing prerequisite, incomplete internship
TOR	Correct course history, correction, secure release
Security	Unauthorized access, role escalation, audit-log integrity
Recovery	Backup restore, transaction recovery, document recovery

37. Reports & Dashboards

Recommended reports include:

- Admission applications by program/status
- Missing admission requirements
- Enrollment by term/program/year/section
- Section capacity and conflicts
- Students with prerequisite deficiencies
- Grade distributions and missing-grade reports

- Academic standing/deficiency reports
- Students at risk of delayed graduation based on objective missing requirements
- Internship/OJT completion
- Graduation candidates and deficiencies
- Clearance status
- TOR requests and releases
- Credential verification activity
- Audit-log reports

38. Document Templates

Build configurable templates for:

- Admission application
- Admission decision notice
- Document checklist
- Enrollment/registration form
- Certificate of Registration (if used)
- Assessment/billing statement
- Academic advising form
- Course withdrawal/add-drop form
- Grade submission form
- Grade correction form
- Transfer-credit evaluation
- Internship/OJT completion
- Thesis/capstone approval
- Graduation application
- Degree audit
- Clearance
- Transcript of Records
- Certificate of graduation/degree
- Other registrar certifications

39. Recommended Development Roadmap

Build in dependency order. Do not begin with the TOR generator alone; the TOR depends on trustworthy upstream academic records.

Stage	Deliverables
1. Discovery	Policies, forms, current workflows, stakeholders, curriculum, grading rules
2. Architecture	Database model, roles, authentication, audit design, integration strategy
3. Master data	Students, programs, courses, curriculum versions, terms, instructors
4. Admissions	Applications, requirements, evaluation, decisions
5. Enrollment	Advising, registration, sections, schedules, add/drop
6. Academic	Attendance, assessments, grades, academic standing
7. Exceptions	Transfers, shifting, leave, readmission, repeats, substitutions
8. Supporting services	Finance, scholarships, internship, research
9. Graduation	Degree audit, clearance, graduation approval
10. Registrar/TOR	Official record, credential generation, verification
11. Security/QA	RBAC, audit logs, backups, testing, privacy controls
12. Deployment	Training, migration, pilot, monitoring, support

40. Minimum Viable Product (MVP)

If development must be staged, the first usable version should include:

- 1 Student master record
- 2 Admissions
- 3 Program/curriculum management
- 4 Academic terms and courses
- 5 Enrollment and sections
- 6 Grades
- 7 Basic academic standing
- 8 Registrar records
- 9 Degree audit
- 10 Graduation records
- 11 TOR generation

Advanced modules—finance, scholarships, OJT, research, alumni, integrations, analytics, and advanced document verification—can follow after the core academic record is stable.

41. Final Development Checklist

Area	Done
Student master record	☐
Admissions workflow	☐
Admission documents	☐

Area	Done
Admission decision	☐
Student number/account	☐
Program/curriculum	☐
Course catalog	☐
Prerequisites/corequisites	☐
Advising	☐
Enrollment	☐
Sections/schedules	☐
Add/drop/withdrawal	☐
Transfer credits	☐
Program shifting	☐
Leave/readmission	☐
Attendance	☐
Assessments	☐
Grades	☐
Grade corrections	☐
Academic standing	☐
Finance/holds	☐
Scholarships	☐
Internship/OJT	☐
Research/thesis/capstone	☐
Graduation audit	☐
Clearance	☐
Degree conferment	☐
TOR	☐
Credential verification	☐
Security/RBAC	☐
Audit logs	☐
Backup/recovery	☐
Testing	☐
Deployment/training	☐

42. Important Implementation Note

This guide is a development blueprint, not a substitute for the university's approved academic policies. Before implementation, the institution should supply the authoritative curriculum, grading policy, enrollment rules, graduation requirements, registrar procedures, document formats, retention rules, privacy requirements, and applicable Philippine higher-education regulations. Configure the system from those approved sources.

End of Guide